

XOM Compound Oil Shock Equity Stress Model

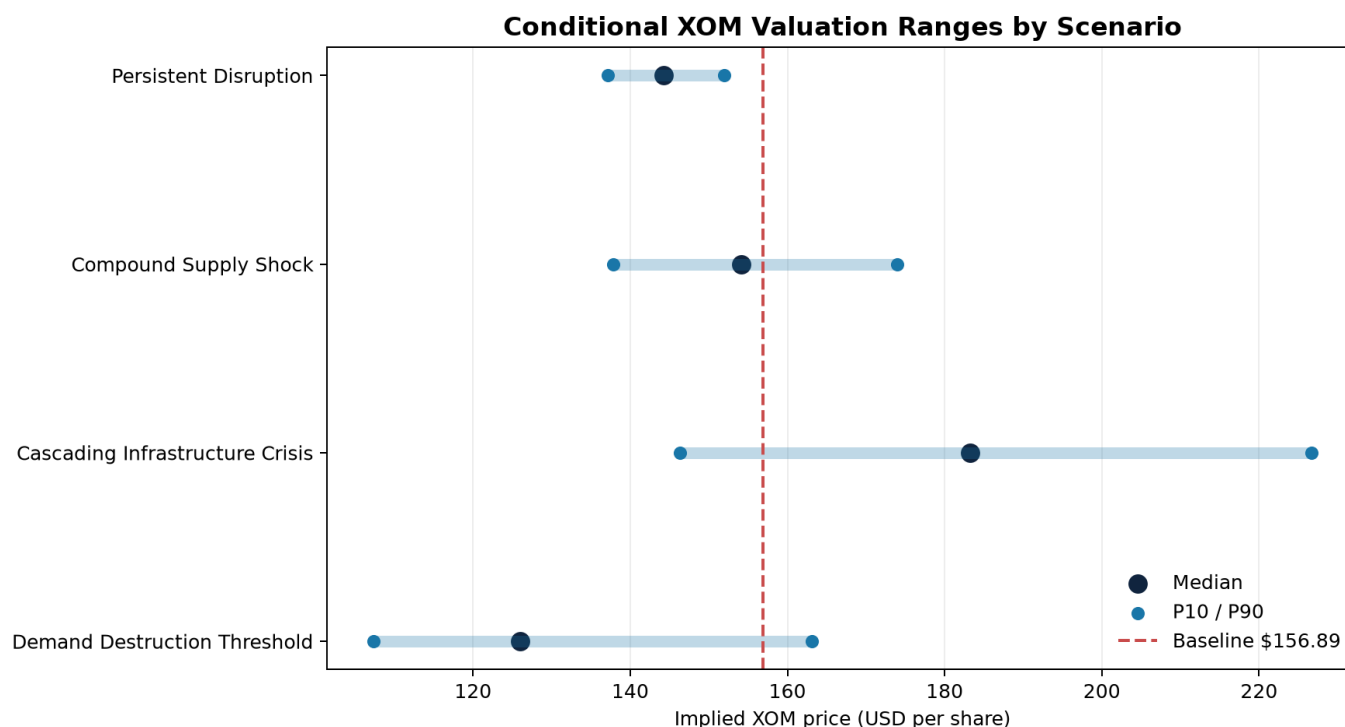
A scenario-based assessment of a prolonged Strait of Hormuz closure combined with attacks on Russian and Black Sea oil infrastructure.

Market baseline: 2026-07-23 | XOM \$156.89 | Brent \$100.69/bbl

Executive judgment

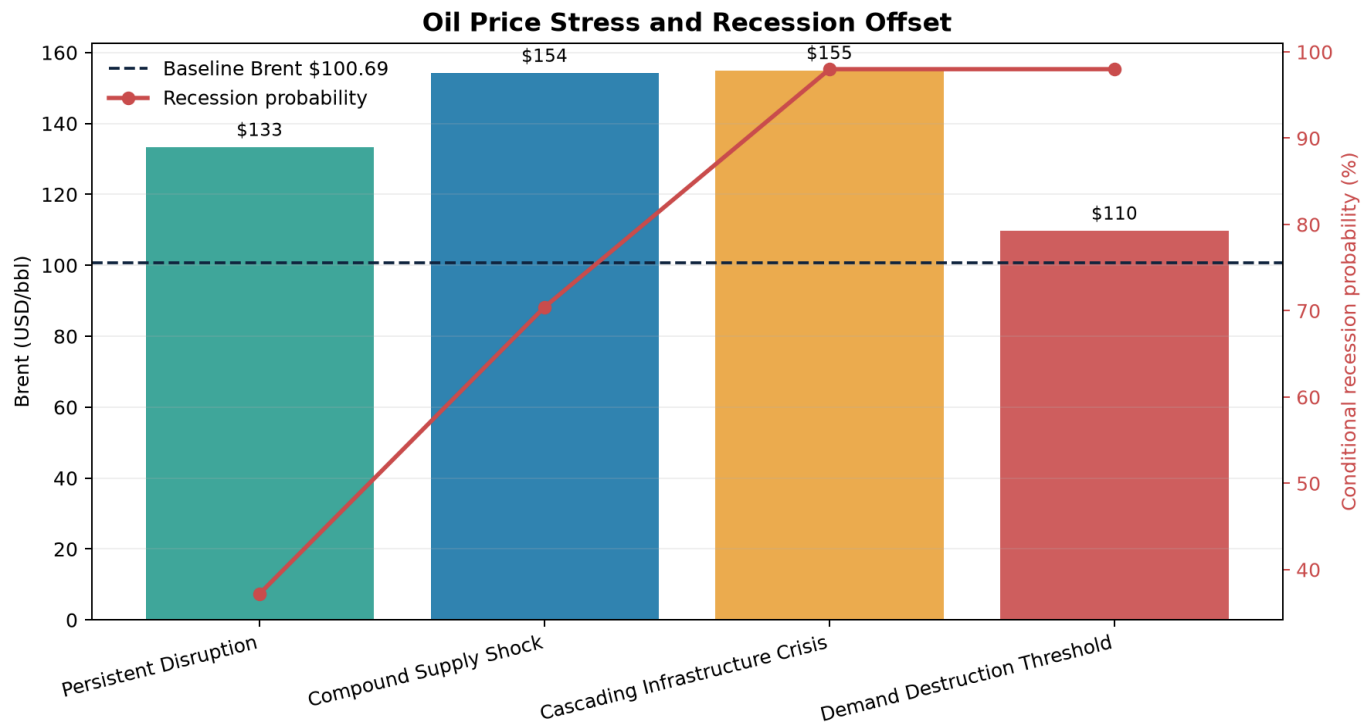
ExxonMobil benefits from higher crude realizations and tighter refining margins during the first phase of a compound supply shock. The benefit is not unlimited. Longer disruptions increase the probability of lost company volumes, demand destruction, recession, working-capital needs, and valuation-multiple compression.

Highest median scenario \$183.28	Demand-threshold median \$125.97	Threshold probability above baseline 14%
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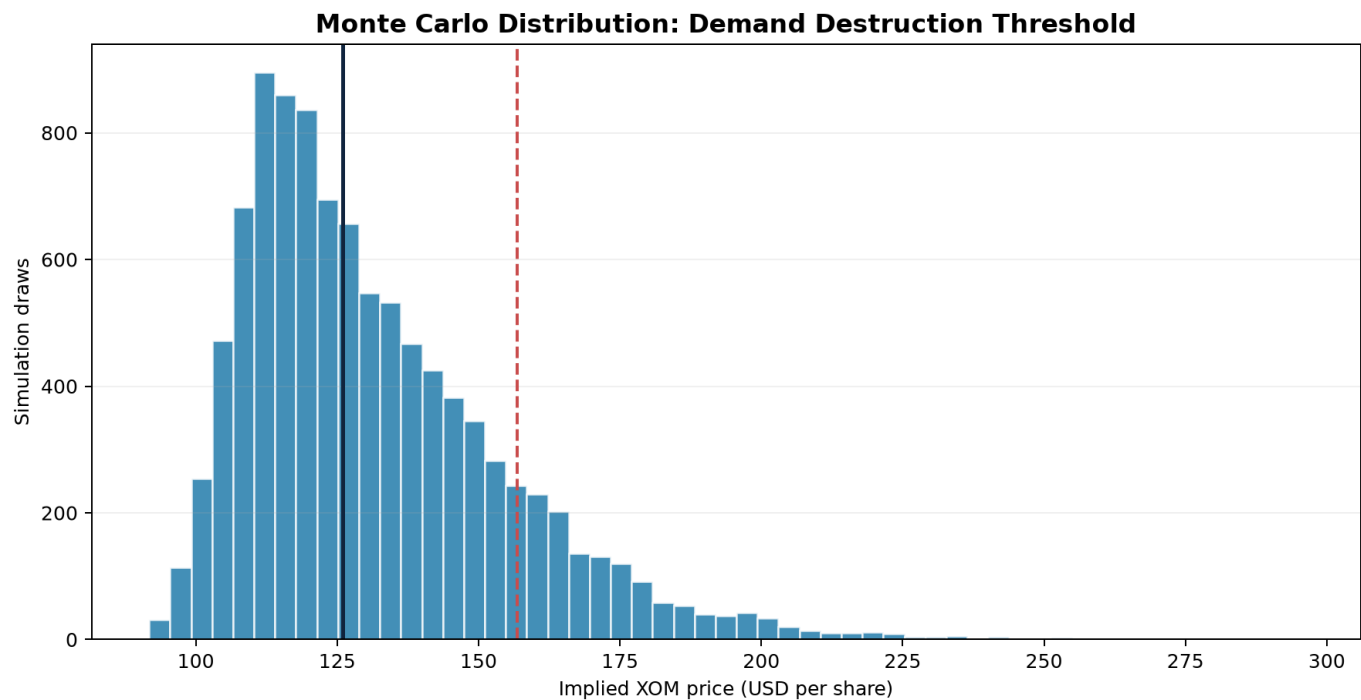
Scenario results

Scenario	Brent	Net income	P10	Median	P90	P(> base)
Persistent Disruption	\$133.27	\$35.2B	\$137.13	\$144.23	\$151.96	2%
Compound Supply Shock	\$154.17	\$47.6B	\$137.87	\$154.11	\$174.00	43%
Cascading Infrastructure Crisis	\$154.95	\$69.5B	\$146.30	\$183.28	\$226.67	81%
Demand Destruction Threshold	\$109.69	\$43.9B	\$107.37	\$125.97	\$163.06	14%



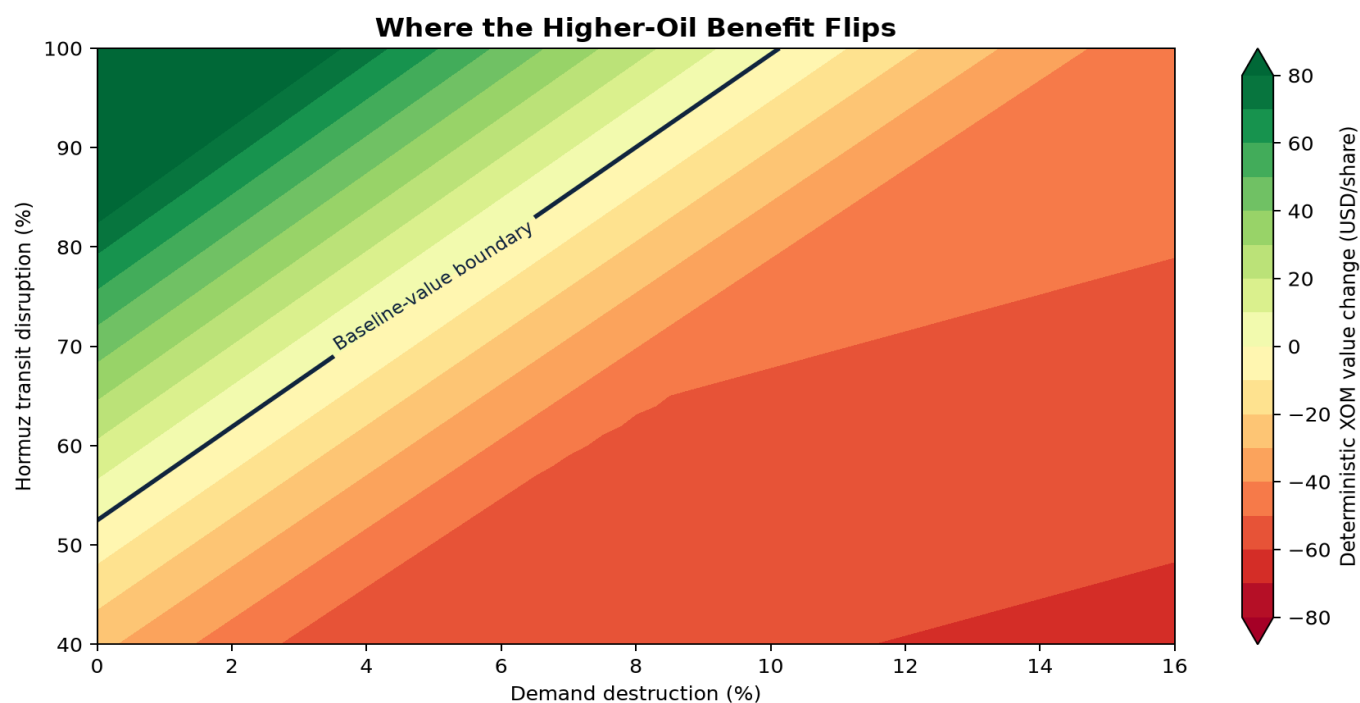
Why the relationship becomes nonlinear

Positive channels	Negative channels
Higher realized crude prices Tighter diesel and gasoline crack spreads Inventory gains Geographic diversification through Permian and Guyana	Tengiz and CPC volume exposure Freight, insurance, and security costs Chemical-margin pressure Demand destruction and recession Equity-risk-premium expansion



Interpretation: the distribution is conditional on the selected scenario. It does not estimate the probability that the geopolitical scenario itself will occur.

Threshold map and method



The dark boundary marks the modeled combination at which the deterministic XOM value crosses the current-share-price baseline. The location of this boundary is conditional on all other cascading scenario assumptions.

Method and limitations

Physical supply layer. Gross Hormuz disruption is reduced by available bypass capacity, policy offsets, and modeled demand response. Russian crude, product, CPC, and other losses are added separately.

Market layer. A convex short-run supply-gap function estimates Brent. Product tightness affects a composite refining crack spread.

Corporate layer. Brent and refining changes flow through Exxon Upstream and Energy Products. Chemical, Specialty, Corporate, volume, working-capital, and security-cost effects are modeled independently.

Valuation layer. The implied share price is an ensemble of earnings-multiple, free-cash-flow-yield, and market-factor estimates. Each scenario uses 10,000 Monte Carlo draws.

Important limitation. Public segment data cannot reproduce Exxon's internal planning model. Sensitivities are transparent research assumptions and should be recalibrated as new filings and event data become available. This is research, not investment advice.

Primary sources

- U.S. EIA - World Oil Transit Chokepoints and Strait of Hormuz analysis.
- ExxonMobil - 2025 results and 2025 Form 10-K.
- Reuters - July 2026 CPC, Tengiz, Russian infrastructure, and crude-price reporting.
- FRED and yfinance - optional live market observations when available.